KAIZEN MATHS

Lesson starter

Trigonometric Differentiation

Chain, Product, and Quotient (Trig Only)

Name _____

S1

Differentiate with respect to x.

$y = \sec(4x + 9)\ \tan(5x + 8)$

S2

Differentiate with respect to x.

$y = \cot(2x^2 - 2x - 3)$

S3

Differentiate with respect to x.

$y = \dfrac{\cos(3x - 8)}{\csc(-7x + 9)}$

S4

Differentiate with respect to x.

$y = (\tan(-8x - 5))^3\ \csc(-9x + 1)$

Show your working

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ALG-C0E7-L2 · S

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KAIZEN MATHS

Exit ticket

Trigonometric Differentiation

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Name _____

E1

Differentiate with respect to x.

$y = \sin(9x + 4)\ \tan(8x - 3)$

E2

Differentiate with respect to x.

$y = \sin(4x^2 - 3x + 4)$

E3

Differentiate with respect to x.

$y = \dfrac{\csc(-2x + 1)}{\tan(6x - 7)}$

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ALG-C0E7-L2 · E

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Chain, Product, and Quotient (Trig Only)

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ALG-C0E7-L2 · E

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Teacher answers & working

Trigonometric Differentiation · Chain, Product, and Quotient (Trig Only)

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S1

Differentiate with respect to x.

$y = \sec(4x + 9)\ \tan(5x + 8)$

Answer:

$\frac{dy}{dx} = \sec(4x + 9)[4\ \tan(4x + 9)\ \tan(5x + 8) + 5\ (\sec(5x + 8))^2]$

Working:

Write the two factors as $f = \sec(4x + 9)$ and $g = \tan(5x + 8)$.

Use product rule: $(fg)\prime = f\prime g + fg\prime$.

$f' = (\sec(4x + 9)\tan(4x + 9))(4)$

$f' = 4\ \sec(4x + 9)\tan(4x + 9)$

$g' = (\sec^2(5x + 8))(5)$

$g' = 5\ \sec^2(5x + 8)$

Substitute into the product rule: $y' = (4\ \sec(4x + 9)\tan(4x + 9))(\tan(5x + 8)) + (\sec(4x + 9))(5\ \sec^2(5x + 8))$.

$y' = 4\ \sec(4x + 9)\tan(4x + 9)\ \tan(5x + 8) + 5\ \sec(4x + 9)\ \sec^2(5x + 8)$

Factor out the common factor $\sec(4x + 9)$.

Factorised final form: $\frac{dy}{dx} = \sec(4x + 9)[4\ \tan(4x + 9)\ \tan(5x + 8) + 5\ (\sec(5x + 8))^2]$

S2

Differentiate with respect to x.

$y = \cot(2x^2 - 2x - 3)$

Answer:

$\frac{dy}{dx} = -\csc^2(2x^2 - 2x - 3)\ (4x - 2)$

Working:

Let $u = 2x^2 - 2x - 3$.

Differentiate the inner function: $u' = 4x - 2$.

Differentiate the outer function: $\frac{d}{du}\cot(u) = -\csc^2(u)$.

$y' = \frac{dy}{du}\frac{du}{dx}$

$y' = (-\csc^2(2x^2 - 2x - 3))(4x - 2)$

$\frac{dy}{dx} = -\csc^2(2x^2 - 2x - 3)\ (4x - 2)$

Teacher answers & working

Trigonometric Differentiation · Chain, Product, and Quotient (Trig Only)

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S3

Differentiate with respect to x.

$$y = \frac{\cos(3x - 8)}{\csc(-7x + 9)}$$

Answer:

$$\frac{dy}{dx} = \frac{-[3\,\sin(3x - 8) + 7\,\cos(3x - 8)\,\cot(-7x + 9)]}{\csc(-7x + 9)}$$

Working:

Identify the numerator and denominator: $u = \cos(3x - 8)$ and $v = \csc(-7x + 9)$.

Use quotient rule: $(u/v)\prime = (u\prime v - uv\prime)/v^2$.

$u' = (-\sin(3x - 8))(3)$

$u' = -3\,\sin(3x - 8)$

$v' = (-\csc(-7x + 9)\cot(-7x + 9))(-7)$

$v' = 7\,\csc(-7x + 9)\cot(-7x + 9)$

Substitute into the quotient rule before simplifying.

$$\frac{dy}{dx} = \frac{(-3\,\sin(3x - 8))\,\csc(-7x + 9) - \cos(3x - 8)\,(7\,\csc(-7x + 9)\cot(-7x + 9))}{(\csc(-7x + 9))^2}$$

Factor out the common factor $-\csc(-7x + 9)$.

$$y' = \frac{-\csc(-7x + 9)[3\,\sin(3x - 8) + 7\,\cos(3x - 8)\,\cot(-7x + 9)]}{(\csc(-7x + 9))^2}$$

Cancel common numerator and denominator factors.

Factorised final form: $\frac{dy}{dx} = \frac{-[3\,\sin(3x - 8) + 7\,\cos(3x - 8)\,\cot(-7x + 9)]}{\csc(-7x + 9)}$

Teacher answers & working

Trigonometric Differentiation · Chain, Product, and Quotient (Trig Only)

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S4

Differentiate with respect to x.

$y = (\tan(-8x - 5))^3 \csc(-9x + 1)$

Answer:

$\frac{dy}{dx} = -3(\tan(-8x - 5))^2 \csc(-9x + 1)[8(\sec(-8x - 5))^2 - 3\tan(-8x - 5)\cot(-9x + 1)]$

Working:

This is a product, and the first factor also needs the chain rule.

Let $f = (\tan(-8x - 5))^3$ and $g = \csc(-9x + 1)$.

$f' = 3(\tan(-8x - 5))^{3-1}(\sec^2(-8x - 5))(-8)$

$f' = -24(\tan(-8x - 5))^2 \sec^2(-8x - 5)$

$g' = (-\csc(-9x + 1)\cot(-9x + 1))(-9)$

$g' = 9\csc(-9x + 1)\cot(-9x + 1)$

Apply product rule: $y' = f'g + fg'$.

$y' = (-24(\tan(-8x - 5))^2 \sec^2(-8x - 5))(\csc(-9x + 1)) + ((\tan(-8x - 5))^3)(9\csc(-9x + 1)\cot(-9x + 1))$

$y' = -24(\tan(-8x - 5))^2 \sec^2(-8x - 5)\csc(-9x + 1) + 9(\tan(-8x - 5))^3 \csc(-9x + 1)\cot(-9x + 1)$

Factor out the common factor $-3(\tan(-8x - 5))^2 \csc(-9x + 1)$.

Factorised final form: $\frac{dy}{dx} = -3(\tan(-8x - 5))^2 \csc(-9x + 1)[8(\sec(-8x - 5))^2 - 3\tan(-8x - 5)\cot(-9x + 1)]$

E1

Differentiate with respect to x.

$y = \sin(9x + 4)\tan(8x - 3)$

Answer:

$\frac{dy}{dx} = 9\cos(9x + 4)\tan(8x - 3) + 8\sin(9x + 4)(\sec(8x - 3))^2$

Working:

Write the two factors as $f = \sin(9x + 4)$ and $g = \tan(8x - 3)$.

Use product rule: $(fg)' = f'g + fg'$.

$f' = (\cos(9x + 4))(9)$

$f' = 9\cos(9x + 4)$

$g' = (\sec^2(8x - 3))(8)$

$g' = 8\sec^2(8x - 3)$

Substitute into the product rule: $y' = (9\cos(9x + 4))(\tan(8x - 3)) + (\sin(9x + 4))(8\sec^2(8x - 3))$.

$y' = 9\cos(9x + 4)\tan(8x - 3) + 8\sin(9x + 4)\sec^2(8x - 3)$

No further common factor remains.

Factorised final form: $\frac{dy}{dx} = 9\cos(9x + 4)\tan(8x - 3) + 8\sin(9x + 4)(\sec(8x - 3))^2$

Teacher answers & working

Trigonometric Differentiation · Chain, Product, and Quotient (Trig Only)

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E2

Differentiate with respect to x.

$y = \sin(4x^2 - 3x + 4)$

Answer:

$\frac{dy}{dx} = \cos(4x^2 - 3x + 4)\,(8x - 3)$

Working:

Let $u = 4x^2 - 3x + 4$.

Differentiate the inner function: $u' = 8x - 3$.

Differentiate the outer function: $\frac{d}{du}\sin(u) = \cos(u)$.

$y' = \frac{dy}{du}\frac{du}{dx}$

$y' = (\cos(4x^2 - 3x + 4))(8x - 3)$

$\frac{dy}{dx} = \cos(4x^2 - 3x + 4)\,(8x - 3)$

E3

Differentiate with respect to x.

$y = \frac{\csc(-2x + 1)}{\tan(6x - 7)}$

Answer:

$\frac{dy}{dx} = \frac{2\,\csc(-2x + 1)[\cot(-2x + 1)\,\tan(6x - 7) - 3\,(\sec(6x - 7))^2]}{(\tan(6x - 7))^2}$

Working:

Identify the numerator and denominator: $u = \csc(-2x + 1)$ and $v = \tan(6x - 7)$.

Use quotient rule: $(u/v)\prime = (u\prime v - uv\prime)/v^2$.

$u' = (-\csc(-2x + 1)\cot(-2x + 1))(-2)$

$u' = 2\,\csc(-2x + 1)\cot(-2x + 1)$

$v' = (\sec^2(6x - 7))(6)$

$v' = 6\,\sec^2(6x - 7)$

Substitute into the quotient rule before simplifying.

$\frac{dy}{dx} = \frac{(2\,\csc(-2x + 1)\cot(-2x + 1))\,\tan(6x - 7) - \csc(-2x + 1)\,(6\,\sec^2(6x - 7))}{(\tan(6x - 7))^2}$

Factor out the common factor $2\,\csc(-2x + 1)$.

Factorised final form: $\frac{dy}{dx} = \frac{2\,\csc(-2x + 1)[\cot(-2x + 1)\,\tan(6x - 7) - 3\,(\sec(6x - 7))^2]}{(\tan(6x - 7))^2}$